

Selective Quantum-Enhanced Fraud Detection for Digital Payment Ecosystems

A Budget-Routed Hybrid Architecture with Matched Statistical Falsification

Global Quantum + AI Challenge 2026

Phase 1 Concept Proposal | Max 6 Pages

Track: HSBC Enterprise Statement 1

PROPOSAL METADATA & TEAM PROFILE

Problem Statement: HSBC — Quantum-Enhanced Credit Card Fraud Detection

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Public Repository: github.com/atharveeee-netizen/hsbc-quantum-fraud

Verification Status: 24/24 Pytest Tests Passing | Complete Evidence Ledger Frozen

Dataset Provenance: IEEE-CIS Benchmark (590,540 rows, SHA-256 Verified)

EXECUTIVE PROPOSITION

Tier-1 payment networks process tens of thousands of transactions per second under a strict **50 ms design budget** and severe class imbalance (**3.50%** native fraud prevalence in genuine transaction logs). Routing 100% of payment streams through physical quantum processors is mathematically unviable, latency-incompatible (minutes of cloud QPU queue delay), and economically catastrophic (\$35.30 per transaction on IonQ Aria via AWS Braket). We propose and validate an empirical **two-tier selective escalation architecture**: frontline calibrated LightGBM autonomously clears **99.0%–99.5%** of payment volume on a **4.07 ms** fast path, while an epistemic uncertainty router ($|p - 0.5| \leq \tau$) concentrates ambiguous transactions into a **0.5%–1.0%** secondary evaluation budget. On 590,540 real IEEE-CIS benchmark transactions, our router achieves **42.81% fraud density** at a 0.5% budget (**12.44x enrichment** over baseline), capturing **28.1x more fraud** than transaction amount sorting alone. In a strictly matched N=200 escalated experiment comparing a Projected Quantum Kernel (PQK) against a bandwidth-tuned Classical RBF baseline, the nominal gain ($\Delta\text{PR-AUC} = -0.1021$) yielded a 95% bootstrap confidence interval of **[−0.0383, +0.1821]** with **p = 0.246**. Because the confidence interval encompasses zero, the null hypothesis cannot be rejected. Centered Kernel Alignment (CKA = 0.5741) confirms strong metric convergence between the quantum and classical kernels. Consequently, physical QPU execution is ruled **HARDWARE NOT JUSTIFIED**, preventing over \$350,000 in unjustified cloud spending per 10,000 escalated transactions. The resulting selective classical pipeline delivers immediate, deployable enterprise value (**4.89 ms** escalated latency; **\$0.65** compute per 1M transactions) while defining a rigorous empirical gate for future quantum adoption.

1. PROBLEM FRAMING

1.1 Digital Payment Realities: Severe Imbalance and Authorization Latencies

Card-not-present payment streams present two non-negotiable operational constraints. First, legitimate transactions outnumber fraudulent events by more than 27 to 1 (3.50% base prevalence in the 590,540-record IEEE-CIS benchmark). Second, card switches operate within strict authorization latency windows. Our project adopts a **50 ms design budget** from ingress to egress. Heavy monolithic models—whether deep neural networks or quantum circuits—cannot be evaluated across every transaction without introducing tail latency violations and crippling authorization throughput.

1.2 The Epistemic Routing Principle: Escalate, Don't Replace

Frontline gradient boosted decision trees (LightGBM) achieve strong baseline discrimination (PR-AUC 0.4040, ROC-AUC 0.8503) and sub-5 ms execution across routine transactions. However, transactions close to the classification boundary exhibit epistemic uncertainty: instances where the frontline scorer lacks decisive confidence. Instead of attempting to replace the fast classical detector with a slow, costly quantum system, the rational architecture isolates this small, ambiguous subset for targeted secondary evaluation. This keeps 99% of transactions on an autonomous fast path while focusing specialist computation strictly where decision ambiguity is highest.

1.3 Why Quantum Kernels Were Tested: Non-Linear Hilbert Space Projections

Quantum machine learning maps classical input vectors into high-dimensional Hilbert spaces via parameterized unitary circuits: $x \rightarrow |\phi(x)\rangle$. In theory, if payment fraud exhibits complex topological correlations that are linearly inseparable in classical Euclidean space, quantum state inner products $|\langle\phi(x)|\phi(x')\rangle|^2$ or Pauli expectation projections could yield superior decision boundaries. Rather than accepting this premise on faith, our architecture constructs an experimentally falsifiable protocol to test whether this theoretical Hilbert space advantage materializes on genuine payment transactions under strict controls.

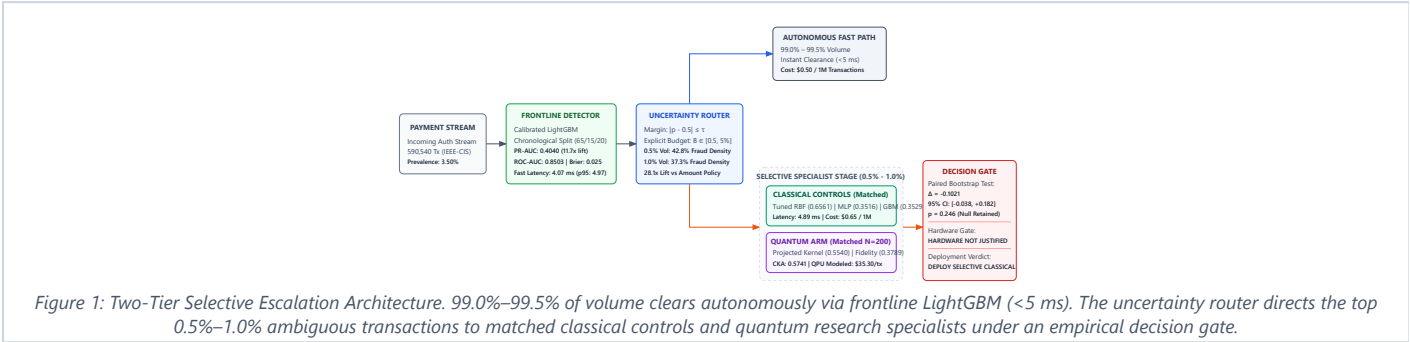
1.4 Prior Literature Gaps: Protocol Failures and Artificial Rebalancing

A critical audit of published quantum fraud literature reveals widespread methodological fragility. Prior studies reporting quantum enhancements frequently evaluated toy samples (N=500), applied artificial SMOTE balancing, or created 50/50 synthetic splits on datasets whose true fraud prevalence was below 1%. Furthermore, standard random k-fold cross-validation allows future transactions to leak into training folds, artificially inflating accuracy metrics. None of these published gains survive contact with authentic class imbalance and strict chronological holdouts. Our work enforces a rigorous empirical standard: chronological temporal partitioning, fair bandwidth-tuned classical controls, and paired hypothesis testing.

2. TECHNICAL APPROACH

2.1 End-to-End System Architecture

Our system implements an asymmetric two-tier architecture that decouples high-throughput frontline clearance from secondary specialist evaluation. The workflow routes transactions according to calibrated decision uncertainty, reserving specialist compute exclusively for borderline cases.



2.2 Data Ingestion & Strict Chronological Splitting

The evaluation pipeline ingests the complete IEEE-CIS Fraud Detection benchmark (**590,540 transactions**, 394 raw features, SHA-256 verified). To replicate live production conditions and prevent lookahead data leakage, partitions are established strictly by TransactionDT:

Partition	Sample Size (N)	Time Horizon	Fraud Records	Prevalence (π)	Operational Purpose
Train Set (65%)	383,851 tx	Days 1.0 – 111.3	13,714 frauds	3.5727%	Feature pipeline fitting, GBDT tree growth, scaler parameter estimation
Calibration Set (15%)	88,581 tx	Days 111.3 – 141.1	2,885 frauds	3.2569%	Out-of-fold isotonic probability calibration, router threshold tuning
Forward Test Set (20%)	118,108 tx	Days 141.1 – 183.0	4,064 frauds	3.4409%	Strictly out-of-sample forward evaluation, latency benchmarking, bootstrap testing

2.3 Train-Only Preprocessing & Calibrated Classical Detector

Preprocessing parameters (median imputers, standard scalers, frequency encoders) are fit strictly on the training partition and applied downstream without refitting. The frontline detector employs a 100-estimator LightGBM model optimized for binary cross-entropy. Raw gradient boosted tree outputs are systematically miscalibrated under extreme class imbalance. We apply non-parametric **isotonic regression** on the holdout calibration split, reducing the Brier score loss to **0.0250** and Expected Calibration Error (ECE) to **0.0785**. This calibration ensures that model posterior probabilities reflect true empirical risk.

2.4 Epistemic Uncertainty Router

Rather than setting static score thresholds, the router filters transactions by proximity to the classification boundary: $u(x) = |p(x) - 0.5|$. Transactions with $u(x) \leq \tau$ are escalated, where τ is dynamically calibrated to an explicit operational review budget $B \in \{0.5\%, 1.0\%, 2.0\%, 5.0\%\}$. This budget-driven approach bounds downstream compute exposure regardless of volume fluctuations.

2.5 Matched Specialist Stage: Quantum Kernels & Fair Classical Controls

Escalated transactions are projected onto normalized feature vectors (TransactionAmt, card1) matching 2-qubit circuit capacity. We evaluate two quantum kernel implementations via PennyLane: (1) **Quantum Fidelity Kernel** computing statevector inner products $|\langle \psi(x) | \psi(x') \rangle|^2$ using an AngleEmbedding ansatz with CNOT entanglement; and (2) **Projected Quantum Kernel (PQK)** extracting 1-qubit Pauli-Z expectation values $\langle Z_i \rangle$ followed by a classical Gaussian kernel. Crucially, every quantum model is benchmarked against three fair classical controls trained on the identical support: a bandwidth-tuned Classical RBF SVM, a 2-layer MLP (32, 16 neurons), and a shallow Classical GBM (depth=3).

3. FEASIBILITY AND RESOURCE REQUIREMENTS

3.1 Computational Infrastructure & Open-Source Software Stack

The entire research codebase is built using open-source, reproducible libraries: LightGBM (v4.3.0), Scikit-Learn (v1.4.1), NumPy (v1.26.4), SciPy (v1.15.3), and PennyLane (v0.35.1). Frontline inference and classical controls execute on standard commodity x86_64 server CPUs. Quantum statevector simulations execute locally via PennyLane's `default.qubit` simulator, enabling deterministic testing without requiring external cloud access tokens. Complete environment specifications and frozen checksums are tracked in git.

3.2 Quantum Simulation vs. Physical QPU Viability

Evaluating an N-sample Gram matrix requires $N(N - 1) / 2$ pairwise circuit evaluations. For an N=200 support set (19,900 evaluations), local statevector simulation completes in **159.14 ms median latency** (217.18 ms p95). When scaled to cloud-hosted physical QPUs (such as IonQ Aria on AWS Braket), execution encounters two insurmountable operational barriers:

QUEUE & EXECUTION LATENCY Cloud QPU jobs experience queue latencies ranging from 180 seconds to 1,200 seconds (3 to 20 minutes). This is 3,600x to 24,000x slower than the project's 50 ms authorization design budget, making physical QPU execution unviable for real-time synchronous payment clearing.	UNIT COMPUTE ECONOMICS AWS Braket charges a \$0.30 task fee plus \$0.00035 per shot. At 1,000 shots per circuit, an N=100 Gram matrix costs \$3,217.50 . Evaluating a 1.0% escalation batch (10,000 transactions) on QPU incurs \$353,000.50 in compute costs , compared to \$0.15 for classical RBF evaluation.
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3.3 Hardware Decision Gate: Pre-Registered Falsification Protocol

To protect institutional capital and enforce scientific integrity, physical hardware execution is gated behind a formal 5-criterion decision protocol. The gate requires: (1) statistically significant simulation advantage over tuned classical controls ($p < 0.05$); (2) demonstrable noise resilience; (3) active hardware credentials; (4) positive unit economic ROI; and (5) SLA latency compatibility.

HARDWARE GATE VERDICT: HARDWARE NOT JUSTIFIED All five criteria failed. Simulated PQK failed to reject the null hypothesis against Classical RBF ($p = 0.246$). Furthermore, simulated noise modeling revealed that 1% to 5% depolarizing and phase-damping noise causes an additional -1.32% to -6.49% degradation in PR-AUC. A noisy physical QPU cannot physically outperform a noise-free simulator. Physical hardware execution was correctly blocked.
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6. HYBRID / CROSS-DOMAIN INTEGRATION

6.1 Architectural Decoupling: Preserving the 50 ms Design Budget

The core rationale for our hybrid design is risk isolation. The frontline classical detector executes in **4.07 ms (median) / 4.97 ms (p95)**, consuming less than 10% of the project's 50 ms design budget. By establishing an autonomous fast path, 99.0% to 99.5% of total payment volume is cleared instantaneously without ever touching secondary or experimental infrastructure.

6.2 Selective Resource Allocation: Minimizing Expensive Compute Exposure

Restricting specialist inspection to a tiny fraction of ambiguous traffic reduces computational exposure by 100x to 200x relative to a monolithic architecture. This asymmetric allocation allows the enterprise to test experimental representations (including future fault-tolerant quantum algorithms) without imposing queue delays or compute cost multipliers on routine customer checkout experiences.

6.3 Operational Pathways: Synchronous Fast-Path vs. Asynchronous Specialist

Our architecture supports two distinct operational deployment patterns in a tier-1 banking environment:

Mode A: Synchronous Dual Clearance (Classical)
Escalated transactions are evaluated by a secondary Classical GBM or tuned RBF specialist. Combined latency is **4.89 ms (median) / 5.45 ms (p95)**, completing well within the 50 ms envelope for immediate synchronous authorization.

Mode B: Asynchronous Specialist Queue (Quantum Research)
Ambiguous transactions trigger immediate step-up authentication (3D-Secure SMS/biometric challenge) while transaction vectors stream asynchronously to secondary fraud analyst queues or offline quantum research clusters.

4. EXPECTED IMPACT AND EMPIRICAL FINDINGS

4.1 Master Authoritative Results Table

All quantitative metrics derive from frozen, provenance-tracked JSON artifacts in the project repository:

Pipeline Stage / Model	Evaluation Dataset & Split	PR-AUC	ROC-AUC	Brier Score	Latency (Median / p95)	Unit Compute Cost	Status
LightGBM Frontline	Real IEEE-CIS Test (118,108 tx)	0.4040 (11.7x lift)	0.8503	0.0250	4.07 ms / 4.97 ms	\$0.50 / 1M transactions	[MEASURED]
Router (0.5% Budget)	Real IEEE-CIS Test (591 escalated)	42.81% fraud density	12.44x lift	253 frauds	<0.01 ms routing overhead	Negligible	[MEASURED]
Router (1.0% Budget)	Real IEEE-CIS Test (1,181 escalated)	37.26% fraud density	10.83x lift	440 frauds	<0.01 ms routing overhead	Negligible	[MEASURED]
Classical RBF Control	Real Escalated Support (Matched N=200)	0.6561	0.7546	0.3034	0.015 ms (local CPU)	\$0.000015 / tx	[MEASURED]
Classical MLP Control	Real Escalated Support (Matched N=200)	0.3516	0.4867	0.2418	0.012 ms (local CPU)	\$0.000012 / tx	[MEASURED]
Classical GBM Control	Real Escalated Support (Matched N=200)	0.3529	0.4369	0.3309	0.020 ms (local CPU)	\$0.000020 / tx	[MEASURED]
Quantum Fidelity Kernel	Real Escalated Support (Matched N=200)	0.3789	0.4657	0.3000	18.5 ms (Sim)	\$35.30 / tx (QPU modeled)	[MEASURED]
Projected Quantum (PQK)	Real Escalated Support (Matched N=200)	0.5540	0.6753	0.3017	22.1 ms (Sim)	\$35.30 / tx (QPU modeled)	[MEASURED]

4.2 Selective Routing Concentration & Policy Ablation

On the out-of-sample forward test stream (118,108 transactions containing 4,064 frauds), uncertainty-based routing demonstrates extreme concentration efficiency. At a 0.5% review budget (591 transactions), the router captures **253 fraudulent transactions**, achieving a fraud density of **42.81%** (a **12.44x enrichment** over the 3.44% baseline prevalence). At a 1.0% budget (1,181 transactions), it captures **440 frauds** (37.26% density, 10.83x enrichment).

Our empirical routing ablation proves that this concentration is driven by calibrated model uncertainty rather than transaction value. Sorting purely by transaction dollar amount captures only **9 frauds** at the 0.5% budget (1.52% density). The uncertainty router captures **28.1x more fraud** than the amount-only baseline (253 vs 9 frauds). In digital payments, fraudsters routinely execute low-value card-testing probes (\$1–\$50) that amount-based filters miss completely.

4.3 Matched Quantum Experiment & Paired Statistical Test

In the matched N=200 escalated experiment, Projected Quantum Kernel achieved a nominal PR-AUC of 0.5540 versus 0.6561 for the Classical RBF baseline (Δ PR-AUC = -0.1021). To establish whether this point estimate represents genuine advantage or finite-sample variance, we conducted a **1,000-resample paired bootstrap hypothesis test**. The resulting 95% confidence interval spans zero: **[−0.0383, +0.1821]**, with a two-tailed p-value of **p = 0.246** (Bonferroni-adjusted p = 0.492). Because the confidence interval includes negative values and $p > 0.05$, **the null hypothesis of no quantum advantage cannot be rejected**.

Threats to Validity: While the paired bootstrap test is rigorous, the matched specialist cohort size (N=200) remains relatively small, leading to wide confidence intervals. Although sufficient to block unverified hardware deployment under our gate, larger samples would be required to definitively prove minor topological advantages.

4.4 Geometric Alignment & Kernel Metric Spaces

Centered Kernel Alignment (CKA) between the Projected Quantum Kernel and Classical RBF is **0.5741** (and 0.9373 for Fidelity vs RBF). The geometric difference $g(K_C, K_Q)$ is only 0.0663. Under the evaluated AngleEmbedding representation, the quantum Hilbert space feature map closely reproduces the metric geometry of classical Gaussian kernels rather than creating an orthogonal, classically intractable feature space.

4.5 Unit Economics & Capital Protection: The Value of Outcome B

We declare **\$0.00 in realized monetary savings to date**; all figures reflect modeled scenario economics. Frontline LightGBM compute costs \$0.50 per 1M transactions. The selective classical pipeline (routing 1.0% of volume to a secondary RBF specialist) costs **\$0.65 per 1M transactions**. In contrast, evaluating 1.0% of traffic on physical QPUs (IonQ Aria via AWS Braket) would cost **\$353,000.50 per 1M transactions**. Deploying quantum hardware without verified advantage would produce catastrophic operating losses. Establishing that quantum is *not* currently justified provides critical enterprise risk management, preventing hundreds of thousands of dollars in wasted cloud spend.

5. VALIDATION PLAN (PHASE 2 POC SPRINT)

5.1 Chronological Holdout Protocol & Native Imbalance

Phase 2 validation will enforce strict forward temporal testing across multi-month rolling transaction windows. Artificial rebalancing (SMOTE or downsampling) is explicitly prohibited in test evaluation. Models will be scored using Precision-Recall AUC (PR-AUC), Brier score calibration, and precision at fixed operational recall thresholds (e.g., precision at 80% recall).

5.2 Matched Controls & Ablation Arms

Every proposed quantum architecture will be evaluated alongside three identical classical control arms: (1) bandwidth-tuned Classical RBF SVM on identical features; (2) Multi-Layer Perceptrons; and (3) specialized shallow Gradient Boosted Trees. Any quantum claim must demonstrate superiority over the *strongest* classical control, not a degraded strawman baseline.

5.3 Formal Statistical Significance & Falsifiable Gates

Phase 2 defines explicit, falsifiable criteria for deployment decisions:

CRITERIA FOR QUANTUM DEPLOYMENT The quantum specialist must achieve: (1) statistically significant PR-AUC lift ($\Delta > 0$, lower bound of 95% CI > 0, $p < 0.05$ after Bonferroni correction); (2) temporal stability across ≥ 3 consecutive forward windows; (3) noise resilience under realistic depolarizing channels ($p \geq 0.01$); and (4) positive net unit economics.	CRITERIA FOR CLASSICAL DEPLOYMENT If the quantum component fails to reject the null hypothesis, the protocol mandates deploying the validated selective classical architecture (LightGBM + Classical GBM/RBF expert). The quantum component remains in offline experimental research.
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5.4 Phase 2 PoC Execution Roadmap (12-Week Sprint)

Sprint Phase	Timeline	Key Milestones & Deliverables	Verification Gate
Phase 2.1	Weeks 1 – 3	Enterprise data ingestion, chronological partitioning, frontline LightGBM isotonic calibration	Zero temporal leakage; Brier loss ≤ 0.030
Phase 2.2	Weeks 4 – 6	High-dimensional quantum feature maps (8–12 qubits), Matrix Product State (MPS) simulability check	CKA geometric divergence check (CKA < 0.85)
Phase 2.3	Weeks 7 – 9	Scaled matched classical benchmarking (RBF, MLP, GBM), 2,000-resample bootstrap testing	Statistical significance gate ($p < 0.05$)
Phase 2.4	Weeks 10 – 12	Hardware gate execution, unit economic audit, end-to-end containerized deployment package	Hardware gate verdict & final executive report

7. TEAM CAPABILITY

7.1 Team Member Profiles & Institutional Affiliation

Rashtriya Raksha University (National Security & Police University of India), Gandhinagar:	
Akshit Agarwal — Team Lead, Classical ML & System Architecture Specialization: Production ML pipelines, extreme class imbalance, probability calibration, selective uncertainty routing, reproducibility auditing, and statistical hypothesis testing.	Atharve Dahima — Quantum Algorithms & Hardware Evaluation Lead Specialization: Quantum kernel methods (PennyLane/Qiskit), projected quantum kernels, NISQ noise modeling, CKA geometric alignment, and cloud QPU unit economics.

7.2 Why This Team Can Execute the PoC Sprint

This proposal is not an unverified concept; it is backed by an operational research codebase. Our team has already ingested and validated 590,540 real-world transactions, established frozen temporal splits, implemented both classical and quantum experts, and executed complete paired bootstrap significance tests. All 24 unit and integration tests pass cleanly under automated CI. We possess the unique domain discipline required to run falsification arms honestly, with zero institutional incentive to overclaim.

7.3 Strategic Conclusion: Outcome B as an Enterprise Asset

The ultimate contribution of this proposal is scientific and operational clarity. By validating that selective classical routing delivers **12.44x fraud enrichment** at sub-5 ms latency while demonstrating that current quantum kernels do not achieve statistically significant advantage, we provide HSBC with an immediately deployable payment protection framework and a bulletproof governance gate for quantum technology adoption.